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BIOSENSIA-DC: A DRUGCLIP-BASED COMPUTATIONAL PROTOTYPE FOR TARGET FISHING APPLIED TO BIORECEPTOR DISCOVERY

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Abstract.
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1. INTRODUCTION

1.1 Biosensors and the bioreceptor-selection problem

A biosensor can be understood as an analytical device that couples a biological recognition mechanism to a signal-transduction mechanism. The biological component is responsible for interacting with the analyte of interest, while the transducer converts that interaction into a measurable signal. In this work, we use the terms *bioreceptor* and *biorecognition element* in this general sense. Common examples include enzymes, antibodies, aptamers, receptors, and other biomolecular structures capable of selective molecular recognition (Eggins, 1996; Banica, 2012; Morales and Halpern, 2018). The nature of this recognition element strongly influences the sensitivity, selectivity, reproducibility, and practical applicability of the resulting biosensor.

Selecting a suitable bioreceptor is therefore one of the central design decisions in biosensor development. This decision is not merely a matter of choosing a molecule that can bind the analyte: the candidate must also be compatible with immobilization, signal generation, stability requirements, sample conditions, and the intended sensing platform. Enzyme-based biosensors, immunosensors, and aptamer-based sensors offer different advantages and limitations, but each case still requires the identification of a recognition mechanism that is selective enough for the target application (Marques and Yamanaka, 2008; Morales and Halpern, 2018).

The broader BioSensIA project addresses this problem as a computational discovery task. Given an analyte molecule, the goal is to identify proteins, protein domains, binding pockets, or other biomolecular structures that may interact with it and therefore deserve further investigation as candidate bioreceptors. In this formulation, the computational system is not expected to replace experimental validation. Rather, it is intended to reduce the search space:

instead of manually inspecting many unrelated biological databases and structural records, the user receives a prioritized list of candidates that can be examined with additional structural, biochemical, and bibliographic evidence. This ranking-oriented view is particularly useful when the biological interactions relevant to sensing are not already obvious from the literature.

BioSensIA-DC is one computational component of this broader workflow. It focuses on the structural target-fishing step: given a query molecule, it ranks candidate protein pockets according to a learned compatibility score. The prototype is based on DrugCLIP, a contrastive representation-learning model originally proposed for virtual screening, and on Uni-Mol-style three-dimensional molecular and pocket representations (Gao et al., 2023; Zhou et al., 2023). BioSensIA-DC uses DrugCLIP as a representation-learning foundation, but extends it with a new target-fishing workflow, data organization strategy, and retrieval implementation. In the original virtual-screening direction, a protein pocket is used as a query to rank molecules. BioSensIA-DC uses the same molecule–pocket compatibility idea in the reverse direction: the molecule becomes the query, and protein pockets become the candidates to be ranked. The resulting ranked list is intended to support bioreceptor discovery, not to provide a definitive proof of binding or biosensor performance.

1.2 Virtual screening, target fishing, and representation learning

Virtual screening and target fishing are closely related computational tasks, but they differ in the direction of the query. In structure-based virtual screening, the starting point is usually a biological target, often represented by a protein binding pocket, and the goal is to rank molecules from a candidate library according to their predicted compatibility with that target. In target fishing, the direction is reversed: the starting point is a molecule, and the goal is to identify proteins, protein domains, binding sites, or other biological targets that may interact with it. In both cases, the practical output is a ranked list, but the biological interpretation is different. Virtual screening asks which molecules may bind a given target; target fishing asks which targets may bind a given molecule.

Several *in silico* target-fishing strategies have been proposed, including ligand-based, receptor-based, pharmacophore-based, docking-based, and machine-learning approaches (Galati et al., 2021). PharmMapper is a representative example of reverse pharmacophore mapping. Given a query compound, it searches for good alignments between the compound and a database of pharmacophore models associated with protein targets (Liu et al., 2010). Its updated version expanded the target pharmacophore database and introduced statistically meaningful ranking scores for the identified targets (Wang et al., 2017). In simplified terms, PharmMapper asks whether the three-dimensional arrangement of chemical features in the query molecule can be fitted to pharmacophore patterns known or inferred for possible targets.

SwissTargetPrediction follows a different principle. It predicts probable targets of a small molecule by comparing the query compound to molecules with known bioactivity data. The method is based on the similarity principle: molecules that are similar in two-dimensional or three-dimensional chemical space are more likely to share biological targets (Daina et al., 2019). Thus, while PharmMapper is primarily organized around reverse matching against pharmacophore models, SwissTargetPrediction is organized around ligand similarity to compounds with experimentally known target annotations. Both tools are valuable references for BioSensIA because they address the same broad question: given a molecule, which biological targets should be considered plausible?

BioSensIA-DC differs from these tools in its representation-learning formulation. Instead of matching a query molecule against an explicit pharmacophore database, as in PharmMapper, or inferring targets from similarity to known ligands, as in SwissTargetPrediction, *BioSensIA-DC uses a learned compatibility model between molecules and protein pockets*. Its starting point is DrugCLIP, a contrastive learning model that represents molecules and protein pockets with separate encoders and trains them so that compatible molecule–pocket pairs are close in a shared latent space (Gao et al., 2023). DrugCLIP formulates virtual screening as a dense-retrieval problem: a pocket is encoded as a query vector, molecules are encoded as candidate vectors, and ranking is performed by comparing these vectors.

The encoders used by DrugCLIP are based on Uni-Mol, a three-dimensional molecular representation-learning framework that incorporates atom types and atomic coordinates (Zhou et al., 2023). A key feature of Uni-Mol is that it is designed to respect the geometry of molecular structures: its internal pair representations are built from interatomic distances, which are invariant to global translations and rotations of the molecule or pocket. At the same time, Uni-Mol includes an equivariant coordinate-prediction mechanism, so that when coordinates are predicted or updated, they transform consistently under rotations and translations of the input structure. In practical terms, this means that the representation is sensitive to the relative three-dimensional arrangement of atoms, but not to arbitrary choices of coordinate-system origin or orientation.

The central idea behind BioSensIA-DC is that the similarity score learned by DrugCLIP can be used in the reverse direction. If DrugCLIP can assign a compatibility score to a molecule–pocket pair for virtual screening, then the same score can be used to rank pockets for a fixed molecule. However, BioSensIA-DC is not merely a direct execution of the original DrugCLIP workflow. DrugCLIP provides the initial molecule–pocket representation framework, but the target-fishing formulation, candidate-pocket preparation, query construction, molecule-to-pocket retrieval pipeline, and initial diagnostic evaluation workflow were conceptualized and implemented from scratch as part of this work. The resulting prototype therefore adapts a virtual-screening representation model to the biosensor-oriented problem of prioritizing candidate bioreceptors for a query analyte.

1.3 Objectives and contributions

This paper describes BioSensIA-DC, a computational prototype for molecule-to-pocket target fishing in the context of bioreceptor discovery. The prototype extends the DrugCLIP representation framework beyond its original virtual-screening workflow. Instead of ranking molecules for a fixed protein pocket, BioSensIA-DC ranks candidate protein pockets for a fixed query molecule, producing a prioritized list of structural candidates for further investigation.

The implemented contributions include the construction of candidate-pocket files from DrugCLIP-compatible data, the construction of query-molecule files from SMILES strings, local DrugCLIP identifiers, PubChem CIDs, or compound names, and the implementation of molecule-to-pocket retrieval. The prototype also generates ranked pocket lists with compatibility scores, and provides an initial benchmarking and diagnostic workflow.

The present work should be read as a computational-prototype and methodology paper. Suitable benchmark validation has not yet been completed. Therefore, no claim is made that the current model is already a reliable target-fishing predictor. Instead, the paper reports the implemented workflow, the conceptual adaptation from virtual screening to target fishing, and the diagnostic issues that must be resolved before predictive performance can be assessed.

2. METHODOLOGY

2.1 Overview of the BioSensIA-DC workflow

BioSensIA-DC implements a molecule-to-pocket retrieval workflow for target fishing. The input is one or more query molecules, and the output is a ranked list of candidate protein pockets. The workflow reuses the molecule and pocket encoders from DrugCLIP, but reverses the practical direction of retrieval: instead of using a pocket to rank molecules, it uses a molecule to rank pockets.

The candidate protein pockets are stored in a library, which can be encoded once and cached for later searches. During retrieval, BioSensIA-DC encodes the query molecule, loads or computes the candidate-pocket embeddings, calculates molecule–pocket similarity scores, and sorts the candidate pockets by these scores. Figure 1 illustrates this workflow.

The resulting ranked list is saved for downstream analysis. Each entry contains a candidate pocket identifier and a compatibility score. This score is used only for ranking; it should not be interpreted as a calibrated binding affinity or as a direct physical measurement.

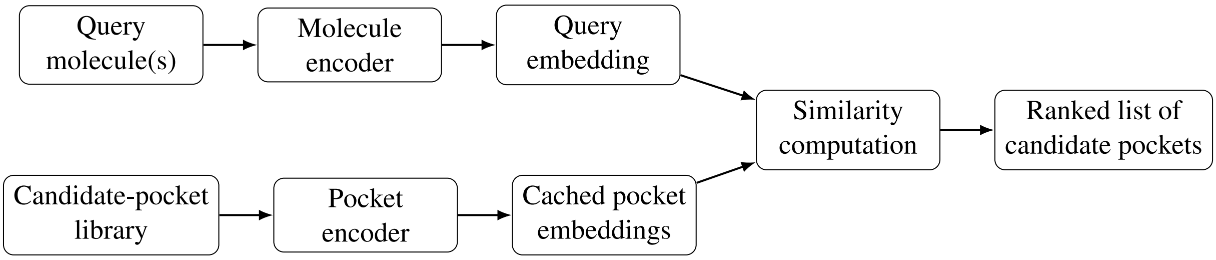

Figure 1: Overview of the BioSensIA-DC target fishing workflow. Query molecules are encoded and compared with cached candidate-pocket embeddings to produce a ranked list of protein pockets.

2.2 Base model: DrugCLIP and Uni-Mol

As Fig. 1 shows, DrugCLIP uses a *two-tower* architecture to represent molecule–pocket compatibility (Gao et al., 2023). One encoder maps each molecule to a latent vector, while a second encoder maps each protein pocket to another latent vector. The compatibility of a molecule–pocket pair is then computed by a similarity score between these two vectors, such as an inner product or cosine similarity.

Both encoders follow the Uni-Mol style of three-dimensional representation learning, in which the input is built from atom types and atomic coordinates rather than only from one-dimensional sequences or two-dimensional molecular graphs (Zhou et al., 2023). This allows the model to represent the local geometry of molecules and protein pockets, which is essential when compatibility depends on the three-dimensional arrangement of atoms.

Mathematically, we have the similarity score $s(p, m)$ defined as the scalar product

$$s(p, m) = g_\phi(x^p)^T \cdot f_\theta(x^m) \tag{1}$$

where p denotes a protein pocket, m denotes a molecule, and x^p and x^m denote their input representations. Furthermore, g_ϕ is the pocket encoder (parameterized by a set ϕ of parameters), and f_θ is the molecule encoder (parameterized by a set θ of parameters). In the current DrugCLIP implementation, both $g_\phi(x^p)$ and $f_\theta(x^m)$ are 128-dimensional vectors.

Figure 2 illustrates how DrugCLIP is trained: both encoders generate embeddings for *positive* pairs (x_i^p, x_i^m) and *negative* pairs $(x_i^p, x_j^m), i \neq j$, each pair containing a pocket and a molecule. Here, "positive" means having evidence of chemical affinity, and "negative" means lacking such evidence. Similarity scores are computed for each pair in the batch and a *loss function* is applied in order to modify the sets of trainable parameters ϕ and θ of the encoders.

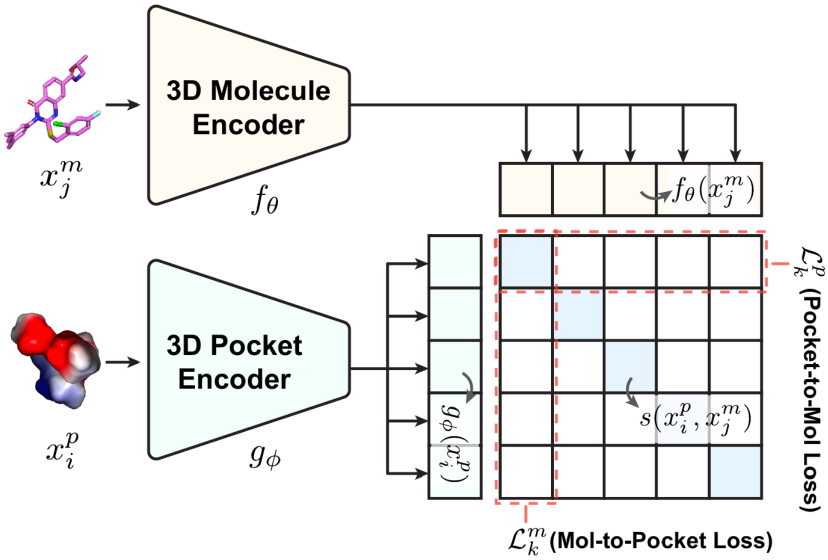

Figure 2: Overview of the DrugCLIP training process. Adapted from (Gao et al., 2023).

Formally, the DrugCLIP loss function has two terms: in a batch of N pairs, the *pocket-to-mol loss* for pocket x_k^p is defined as

$$\mathcal{L}_k^p(x_k^p, \{x_i^m\}_{i=1}^N) = -\log \frac{\exp(s(x_k^p, x_k^m)/\tau)}{\sum_i \exp(s(x_k^p, x_i^m)/\tau)} \tag{2}$$

and the *mol-to-pocket loss* for molecule x_k^m is defined as

$$\mathcal{L}_k^m = (x_k^m, \{x_i^p\}_{i=1}^N) = -\log \frac{\exp(s(x_k^p, x_k^m)/\tau)}{\sum_i \exp(s(x_i^p, x_k^m)/\tau)} \tag{3}$$

where $s(\cdot, \cdot)$ is defined in Eq.(1) and τ is a hyperparameter (called *temperature*) that controls the smoothness of the softmax distribution. The final loss for a batch (which training seeks to minimize) is then

$$\mathcal{L} = \frac{1}{2} \sum_{k=1}^{N} (\mathcal{L}_k^p + \mathcal{L}_k^m) \tag{4}$$

The definitions above are important for the following section, where we describe the modifications we applied to the training procedure in order to fine-tune the model for the target-fishing problem.

2.3 Fine-tuning formulation for target fishing

- Explain that the original DrugCLIP contrastive loss is naturally aligned with virtual screening, but not necessarily optimized for target fishing.

- Describe the issue with repeated ligands:
  - the same molecule can appear in several positive protein-ligand complexes;
  - those complexes may correspond to different structures, homologous proteins, isoforms, or related binding sites;
  - in target fishing, all known pockets for the same molecule should be treated as positives;
  - the original diagonal-only contrastive loss may treat some known positives as implicit negatives, depending on batch composition.
- Define the set of known positive pockets for a molecule:

$$P^+(m) = \{p : (p, m) \in D\} \tag{5}$$

- Explain that D denotes the available set of observed positive pocket-molecule pairs.
- Define a positive-pair mask for a batch:

$$Y_{ij} = \begin{cases} 1, & \text{if } p_j \in P^+(m_i) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

- Define the multi-positive molecule-to-pocket loss:

$$L_i^{m \to p} = -\log \frac{\sum_j Y_{ij} \exp(S_{ij}/\tau)}{\sum_j \exp(S_{ij}/\tau)} \tag{7}$$

- Explain that $S_{ij} = s(p_j, m_i)$ is the similarity between pocket p_j and molecule m_i, and that τ is the contrastive temperature.
- Explain the intuition: for each query molecule, the loss increases the total softmax probability assigned to all known positive pockets in the batch.
- Present fine-tuning variants to be discussed:
  - freeze both encoders and train only projection layers;
  - train projection layers plus the temperature parameter;
  - unfreeze upper encoder layers;
  - train the full model with a smaller learning rate;
  - compare molecule-to-pocket loss alone with a symmetric loss that also includes pocket-to-molecule terms.
- State that the current implementation and experiments are preliminary and that the poor validation results do not yet support a performance claim.

2.4 Benchmarking and diagnostic evaluation

- Explain that benchmarking target fishing is difficult because a query molecule may have several true targets, and missing pairs are not necessarily true negatives.
- Describe the implemented benchmark workflow:
  - construct an explicit table of positive molecule-pocket pairs;
  - build query-molecule LMDB files;
  - build candidate-pocket LMDB files;
  - run molecule-to-pocket retrieval;
  - compare ranked lists against known positives;
  - compute ranking metrics.
- List metrics to be discussed:
  - top-k accuracy;
  - recall@k;
  - mean reciprocal rank;
  - rank of the first known positive;
  - enrichment-style metrics for early retrieval;
  - AUROC only with caution, because it may be less informative when early ranking is the main objective.
- Explain that preliminary results are poor for both the fine-tuned BioSensIA-DC model and the original DrugCLIP model in the current evaluation setup.
- Explain that this suggests the need for diagnostic work before any strong conclusion:
  - verify data splits;
  - verify checkpoint compatibility;
  - verify LMDB contents;
  - verify atom dictionaries;
  - verify cached embeddings;
  - verify whether candidate pockets match the positive-pair table;
  - verify whether the benchmark task is consistent with the model's training assumptions.

2.5 Computational environment and reproducibility

- Describe the computational environment used for development:
  - GNU/Linux;
  - Python 3.11;
  - PyTorch with CUDA;
  - RDKit;
  - Uni-Core;
  - DrugCLIP;
  - LMDB.
- Explain the use of `uv` for Python environment management.
- Explain that a GPU is required by inherited constraints in the original DrugCLIP code path.
- Discuss technical issues encountered:
  - compiling Uni-Core;
  - CUDA extension compatibility;
  - C++ standard-library compatibility;
  - GPU memory limits;
  - long-running embedding precomputation;
  - embedding-cache invalidation.
- Explain why these details matter: they affect reproducibility and practical adoption of 3D deep-learning models in computational biosensor research.

3. RESULTS AND DISCUSSION

3.1 Implemented software artifacts

- Present the public repository as the main concrete result of this stage.
- List implemented artifacts:
  - query-molecule LMDB generation;
  - candidate-pocket LMDB generation;

- molecule-to-pocket retrieval;
- ranked-pocket output;
- preliminary benchmark code;
- installation and execution documentation.

- Include a table with columns such as:
  - file or module name;
  - purpose;
  - input;
  - output;
  - current status.
- Discuss how these artifacts convert DrugCLIP from a virtual-screening-only workflow into a prototype that can also run target-fishing queries.

3.2 Qualitative query example with okadaic acid

- Present okadaic acid as a qualitative sanity-check query.
- Explain the biological expectation: PP2A-related pockets should be considered relevant because PP2A is a known okadaic-acid target.
- Include a planned table with:
  - rank;
  - PDB identifier;
  - score;
  - protein or pocket annotation;
  - whether the result is known, plausible, unclear, or likely irrelevant.
- Discuss whether PP2A-associated structures appear near the top of the ranking.
- State clearly that this qualitative example is not a benchmark. It is only a sanity check and an illustration of the workflow.

3.3 Preliminary quantitative results

- Report that the quantitative results obtained so far are unsatisfactory.
- Separate two observations:
  - the fine-tuned BioSensIA-DC model currently performs poorly;
  - the original DrugCLIP model also performs poorly under the current local benchmark workflow.
- Avoid claiming improvement over DrugCLIP.
- Present the results as a diagnostic table, not as a final performance table. Suggested columns:
  - model;
  - checkpoint;
  - query set;
  - candidate-pocket set;
  - metric values;
  - diagnostic comment.
- Explain that the current evidence points to unresolved issues in reproduction, data handling, benchmark construction, or model adaptation.

3.4 Possible causes of the poor benchmark results

- Discuss possible data problems:
  - mismatch between training, validation, and candidate LMDB files;
  - differences between the downloaded data and the data used in the original DrugCLIP paper;
  - duplicated PDB identifiers with different structures;
  - ligand-identity inconsistencies;
  - salt, tautomer, stereochemistry, or conformer inconsistencies.
- Discuss possible implementation or protocol problems:
  - wrong or incompatible checkpoint;
  - stale cached embeddings;

- incorrect correspondence between pocket identifiers and benchmark positives;
- candidate libraries that do not contain the expected positives;
- metrics computed over an inappropriate candidate set;
- target-fishing evaluation derived from data originally organized for virtual screening.

• Discuss possible modeling limitations:

- DrugCLIP was primarily motivated by virtual screening;
- target fishing is multi-positive by nature;
- unobserved pairs are not reliable true negatives;
- the similarity score is not physically calibrated;
- fine-tuning projection layers alone may be insufficient.

• Conclude that the next step must be a systematic audit before interpreting the model's predictive value.

3.5 Relationship to existing target-fishing tools

• Compare BioSensIA-DC conceptually with PharmMapper:

- PharmMapper performs reverse pharmacophore mapping;
- BioSensIA-DC performs learned molecule-pocket representation matching;
- both produce ranked target-candidate lists.

• Compare BioSensIA-DC conceptually with SwissTargetPrediction:

- SwissTargetPrediction relies on similarity to known ligands;
- BioSensIA-DC relies on learned 3D compatibility between molecules and pockets;
- the two approaches may be complementary.

• Discuss how future versions of BioSensIA could combine rankings from multiple methods:

- rank aggregation;
- consensus among tools;
- grouping by protein family;
- docking-based follow-up;
- literature-based validation.

3.6 Relevance to the BioSensIA project

- Explain that BioSensIA-DC is one module in a larger computational pipeline for biosensor-oriented bioreceptor discovery.
- Relate the prototype to planned BioSensIA components:
  - UniProt and Gene Ontology queries;
  - domain and family analysis;
  - PDB and AlphaFold structural data;
  - pocket prediction;
  - docking and pharmacophore analysis;
  - evidence aggregation for candidate bioreceptors.
- Explain that even before full validation, BioSensIA-DC provides an operational way to turn a molecule-centered question into a ranked list of structural candidates.
- Emphasize that the current results motivate further validation rather than immediate deployment.

4. CONCLUSIONS

4.1 Summary of contributions

- Summarize that the paper presented BioSensIA-DC, a DrugCLIP-based computational prototype for molecule-to-pocket target fishing.
- Highlight the main engineering contribution: implementing the reverse retrieval direction needed to rank candidate pockets for a query molecule.
- Highlight that the prototype includes data preparation, retrieval execution, ranked output generation, and preliminary evaluation tools.
- Highlight that the implementation is publicly available and intended to support reproducible development.

4.2 Limitations

- State that the current model has not yet been validated as a reliable predictor.
- State that preliminary benchmark results are poor for both the fine-tuned model and the original DrugCLIP model in the current local evaluation setup.
- Explain that these results may reflect unresolved problems in reproduction, data consistency, benchmark construction, or the adaptation of DrugCLIP to target fishing.
- State that no claim of improved predictive performance is made in the present paper.

4.3 Future work

- Reproduce the original DrugCLIP benchmarks more carefully before interpreting target-fishing results.
- Audit checkpoints, LMDB files, atom dictionaries, cached embeddings, and benchmark-positive definitions.
- Build a target-fishing benchmark with rigorous train-test separation and clear ligand-identity policies.
- Evaluate multi-conformer query strategies, including maximum, mean, median, quantile, and log-sum-exp aggregation.
- Expand the candidate-pocket library using PDB, BioLiP, sc-PDB, AlphaFold structures, and pocket-prediction tools.
- Compare BioSensIA-DC rankings with PharmMapper, SwissTargetPrediction, and docking-based validation.
- Integrate BioSensIA-DC into a broader BioSensIA interface for biosensor-oriented candidate discovery.

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- Include the required generative-AI disclosure. Suggested wording:
  - The author used ChatGPT (OpenAI) to assist with manuscript organization, language revision, and discussion of alternative formulations. All scientific content, interpretation of results, and conclusions remain the responsibility of the author.

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